Circuit investigation

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use measurements to test a circuit model safely.

SUCCESS CRITERIA - I can...

1. Draw and predict before powering a circuit.
2. Place ammeters in series and voltmeters in parallel.
3. Compare measured and predicted values using evidence.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A circuit has 3 identical series resistors. Add two more at unchanged ideal supply voltage. Compare new current with the original.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: V = IR.
Use approved low voltage. Ammeter in series; voltmeter across a component. Disconnect before rewiring; never put an ammeter across the supply.

Worked reasoning: Supplied measurements show V=2, 4, 6 V and I=0.2, 0.4, 0.6 A. Each V/I is 10 ohms. This supports an ohmic model over the measured range.

Your turn: Plot V against I and mark the slope. State why the data do not prove the model at every possible temperature or voltage.

Marzano check - circle: ① ② ③

03 Represent and interpret

On a resistor circuit sketch, add an ammeter to measure loop current and a voltmeter to measure the resistor's drop.



Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

04 Solve and justify

A supplied trial gives V=5.8 V and I=.46 A for a nominal 12 Ω resistor. Find measured V/I and relative deviation from 12 Ω. Name a plausible uncertainty source.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Analyze the reasoning

A reading disagrees with your prediction. List two checks before replacing the model or deleting data.

Marzano check - circle: 1 2 3

06 Choose a method

An ideal blackbody has peak wavelength 580 nm. Use $\lambda_{max}\,T = 2.90 \times 10^{6}$ nm·K. Estimate temperature.

Marzano check - circle: 1 2 3

07 Exit check

Design a three-point V-I test. State what to control and what pattern would support constant resistance.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Two readings are 3.0±0.1 V and 0.30±0.02 A. Find central resistance and worst-case bounds.

Marzano check - circle: 1 2 3